INCH-POUND

AN840 Rev 11
w/AMENDMENT 1
18 November 2024
SUPERSEDING
AN840 Rev 11
1 October 2014

DETAIL SPECIFICATION SHEET

ADAPTER, STRAIGHT, PIPE TO HOSE

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5183.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843/2.

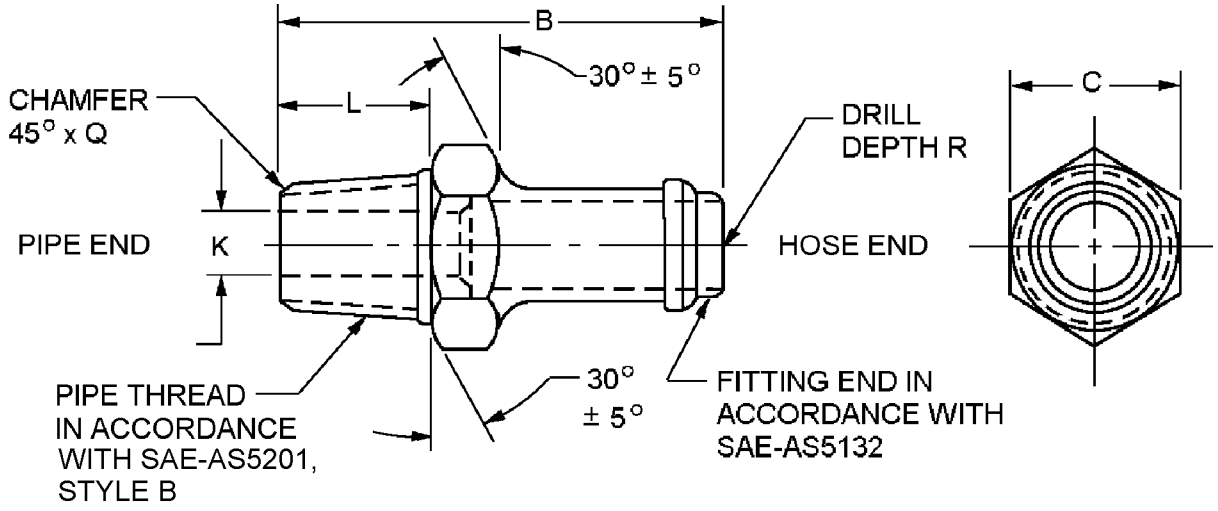


FIGURE 1. Adapter, straight, dimensions and configuration.

AMSC N/A

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Dash number	Hose ID inches (mm)	Pipe thread SAE-AS5201, style B	B inches (mm) ±.031 (0.79)	C inches (mm)	
4	.250 (6.35)	1/8-27 ANPT	2.078 (52.78)	.437 (11.10)	+.003 (0.08) -.004 (0.10)
6	.375 (9.53)	1/4-18 ANPT	2.344 (59.54)	.625 (15.88)	±.004 (0.10)
8	.500 (12.70)	3/8-18 ANPT	2.375 (60.33)	.750 (19.05)	±.004 (0.10)
9	.625 (15.88)	3/8-18 ANPT	2.375 (60.33)	.875 (22.23)	±.004 (0.10)
10	.625 (15.88)	1/2-14 ANPT	2.578 (65.48)	.937 (23.80)	±.004 (0.10)
11	.750 (19.05)	3/8-18 ANPT	2.406 (61.11)	1.000 (25.40)	±.004 (0.10)
12	.750 (19.05)	3/4-14 ANPT	2.625 (66.68)	1.125 (28.58)	±.005 (0.13)
13	1.000 (25.40)	3/8-18 ANPT	2.625 (66.68)	1.250 (31.75)	±.005 (0.13)
16	1.000 (25.40)	3/4-14 ANPT	2.625 (66.68)	1.250 (31.75)	±.005 (0.13)
17	1.000 (25.40)	1-11.5 ANPT	2.844 (72.24)	1.375 (34.93)	±.005 (0.13)
20	1.250 (31.75)	1/ 1/4-11.5 ANPT	2.891 (73.43)	1.750 (44.45)	±.016 (0.41)
21	1.250 (31.75)	1-11.5 ANPT	2.875 (73.03)	1.500 (38.10)	±.005 (0.13)
24	1.500 (38.10)	1 1/2-11.5 ANPT	2.938 (74.63)	2.000 (50.80)	±.016 (0.41)
25	1.500 (38.10)	1 1/4-11.5 ANPT	2.906 (73.81)	1.750 (44.45)	±.016 (0.41)

Dash number	K dia. ±.003 (0.08) inches (mm) (See note 6)	L inches (mm)	Q inches (mm)	R inches (mm)
4	---	.391 (9.93)	.031 (0.79)	---
6	---	.594 (15.09)	.047 (1.19)	---
8	---	.594 (15.09)	.047 (1.19)	---
9	.403 (10.24)	.594 (15.09)	.047 (1.19)	1.625 (41.28)
10	---	.766 (19.46)	.063 (1.60)	---
11	.403 (10.24)	.594 (15.09)	.047 (1.19)	1.625 (41.28)
12	---	.781 (19.84)	.063 (1.60)	---
13	.403 (10.24)	.719 (18.26)	.047 (1.19)	1.625 (41.28)
16	.717 (18.21) min .736 (18.69) max (see note 7)	.781 (19.84)	.063 (1.60)	1.656 (42.06)
17	---	.969 (24.61)	.078 (1.98)	---
20	---	.984 (24.99)	.078 (1.98)	---
21	.887 (22.53)	.969 (24.61)	.078 (1.98)	1.656 (42.06)
24	---	1.000 (25.40)	.078 (1.98)	---
25	1.091 (27.71)	.969 (24.61)	.078 (1.98)	1.656 (42.06)

FIGURE 1. Adapter, straight, dimensions and configuration - Continued.

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Dash number	Weight max lbs (kg)		
	Copper alloy	Al alloy	Steel or CRES
4	.039 (0.018)	.013 (0.006)	.038 (0.02)
6	.069 (0.031)	.023 (0.010)	.067 (0.03)
8	.099 (0.044)	.033 (0.015)	.097 (0.04)
9	.126 (0.057)	.042 (0.019)	.123 (0.06)
10	.171 (0.078)	.057 (0.026)	.167 (0.08)
11	.159 (0.072)	.053 (0.024)	.156 (0.07)
12	.258 (0.117)	.086 (0.039)	.252 (0.11)
13	.258 (0.117)	.086 (0.039)	.252 (0.11)
16	.270 (0.122)	.090 (0.041)	.264 (0.12)
17	.381 (0.173)	.127 (0.057)	.373 (0.17)
20	.681 (0.309)	.227 (0.103)	.666 (0.30)
21	.471 (0.214)	.157 (0.071)	.461 (0.21)
24	.825 (0.374)	.275 (0.125)	.807 (0.37)
25	.705 (0.320)	.235 (0.107)	.689 (0.31)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers.
5. Machined surfaces shall be finished to 125µin Ra, forged surfaces shall be 250µin Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. Dimensions not specified are in accordance with SAE-AS5201.
7. These dimensions are limits and specified tolerances do not apply.
8. For design features purposes, this standard takes precedence over documents referenced herein.

FIGURE 1. Adapter, straight, dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

Installation shall be in accordance with MS21344.

Materials and finishes shall be in accordance with SAE-AS4843/2, see table I for material code.

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TABLE I. Material and finish code letters.

Material and finish code	Material	Protective chemical finish 3/ 4/
B 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	No finish.
BC 1/	Copper alloy, type 377 forging in accordance with ASTM B124/B124M or half hard forging or bar in accordance with ASTM B138/B138M or bar in accordance with SAE-AMS4614.	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 3.
D 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625 , type II, class 2, dye blue, duplex seal in accordance with procurement specification.
DV 2/	Type 2014-T6 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4133, or type 2024-T6 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6, or type 2024-T851 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/6 or SAE-AMS4339.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.
J	Type 304 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5639.	Passivate in accordance with SAE-AMS2700, type VI or VII.
K	Type 316 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5648.	Passivate in accordance with SAE-AMS2700, type VI or VII.

See notes at end of table.

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TABLE I. Material and finish code letters - Continued.

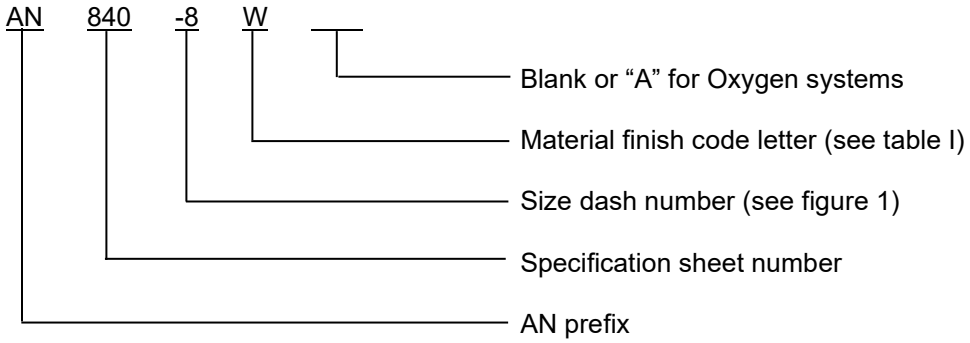
Material code	Material	Protective chemical finish 3/
R	Type 321 corrosion resistant steel forging or bar in accordance with SAE-AMS-QQ-S-763 or SAE-AMS5645.	Passivate in accordance with SAE-AMS2700, type VI or VII.
Dash (-)	Steel type 4130 in accordance with SAE-AMS-S-6758, or SAE-AMS-6370 or type 4140 in accordance with SAE-AMS6382	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 2
W 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	Anodize in accordance with SAE-AMS2472 or MIL-PRF-8625 , type II, class 2, dye brown similar to color 10080 in accordance with SAE AMS-STD-595, duplex seal in accordance with procurement specification.
WV 2/	Type 7075-T73 aluminum alloy forging in accordance with SAE-AMS-QQ-A-367 or SAE-AMS4141, or type 7075-T73 aluminum alloy bar in accordance with SAE-AMS-QQ-A-225/9, in accordance with type 7075-T7351 Aluminum alloy bar in accordance with SAE-AMS4124.	High purity aluminum in accordance with MIL-DTL-83488, class 3, type II with maximum coating thickness of .0005 inch. Glass bead peen pressure shall be 25 psi (1.72 bar) maximum.

1/ Material code was dash on previous revisions, changed to agree with SAE-ARP1590.
2/ Aluminum code D is inactivated, use code W.
3/ Cadmium shall not be used in potable water systems.

Oxygen systems for aerospace, shipboard, and ground support equipment. Parts for use on oxygen systems shall be identified in the PIN as code “A” and shall be furnished cleaned, packaged, and labeled in accordance with SAE-AS611 to a process approved by the user.

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Part or Identifying Number (PIN): The PIN consists of the prefix “AN”, specification sheet number, dash number for hose and flared tube size, material finish code letter(s) and blank or “A” for oxygen cleaning requirements. Unassigned PIN’s shall not be used.



PIN example: AN840-8W indicates an adapter, straight, pipe to hose, .6250 inch (0.159 mm), aluminum alloy 7075-T73.

Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

AN PIN to superseding AN PIN see table II.

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TABLE II. Superseding AN PIN.

AN PIN Inactive part	Hose Size	Pipe Size	Superseding AN PIN Inactive part replacement
AN840-4	0.250	0.125	AN840-4B
AN840-6	0.375	0.250	AN840-6B
AN840-8	0.500	0.375	AN840-8B
AN840-9	0.625	0.375	AN840-9B
AN840-10	0.625	0.500	AN840-10B
AN840-11	0.500	0.375	AN840-11B
AN840-12	0.750	0.750	AN840-12B
AN840-13	1.000	0.375	AN840-13B
AN840-16	1.000	0.750	AN840-16B
AN840-17	0.750	1.000	AN840-17B
AN840-20	1.250	1.250	AN840-20B
AN840-21	1.250	1.000	AN840-21B
AN840-24	1.500	1.500	AN840-24B
AN840-25	1.500	1.250	AN840-25B
AN840-4S	0.250	0.125	AN840-4-
AN840-6S	0.375	0.250	AN840-6-
AN840-8S	0.500	0.375	AN840-8-
AN840-9S	0.625	0.375	AN840-9-
AN840-10S	0.625	0.500	AN840-10-
AN840-11S	0.500	0.375	AN840-11-
AN840-12S	0.750	0.750	AN840-12-
AN840-13S	1.000	0.375	AN840-13-
AN840-16S	1.000	0.750	AN840-16-
AN840-17S	0.750	1.000	AN840-17-
AN840-20S	1.250	1.250	AN840-20-
AN840-21S	1.250	1.000	AN840-21-
AN840-24S	1.500	1.500	AN840-24-
AN840-25S	1.500	1.250	AN840-25-

Table III provides a detailed cross-reference of AN840 PINs and replacement SAE-AS5183 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

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TABLE III. Cross-reference data. 1/

Old AN PIN	Current AN PIN Inactive for new design	Tube Size	Pipe Size	Replacement AS PIN	Replacement AS PIN For new design
AN840-4		.250	.188	AS5183B0402	
AN840-4D	AN840-4W	.250	.188	AS5183W0402	AS5183W0402
AN840-4DV	AN840-4WV	.250	.188	None	
None	AN840-4R	.250	.188	AS5183R0402	AS5183R0402
AN840-4S		.250	.188	AS5183-0402	AS5183-0402
AN840-4T		.250	.188	None	
AN840-4W		.250	.188	AS5183W0402	
AN840WV		.250	.188	None	
AN840-6		.375	.250	AS5183B0604	
AN840-6D	AN840-6W	.375	.250	AS5183W0604	AS5183B0604
AN840-6DV	AN840-6WV	.375	.250	None	
None	AN840-6R	.375	.250	AS5183R0604	AS5183R0604
AN840-6S		.375	.250	AS5183-0604	AS5183-0604
AN840-6T		.375	.250	None	
AN840-6W		.375	.250	AS5183W0604	
AN840-6wv		.375	.250	None	
AN840-8		.500	.375	AS5183B0806	
AN840-8D	AN840-8W	.500	.375	AS5183W0806	AS5183W0806
AN840-8DV	AN840-8WV	.500	.375	None	
None	AN840-8R	.500	.375	AS5183R0806	AS5183R0806
AN840-8S		.500	.375	AS5183-0806	AS5183-0806
AN840-8T		.500	.375	None	
AN840-8W		.500	.375	AS5183W0806	
AN840-8WV		.500	.375	None	
AN840-9		.625	.375	AS5183B1006	
AN840-9D	AN840-9W	.625	.375	AS5183W1006	AS5183W1006
AN840-9DV	AN840-9WV	.625	.375	None	
None	AN840-9R	.625	.375	AS5183R1006	AS5183R1006
AN840-9S		.625	.375	AS5183-1006	AS5183-1006
AN840-9T		.625	.375	None	
AN840-9W		.625	.375	AS5183W1006	
AN840-9WV		.625	.375	None	None
AN840-10		.625	.500	AS5183B1008	
AN840-10D	AN840-10W	.625	.500	AS5183W1008	AS5183W1008
AN840-10DV	AN840-10WV	.625	.500		
None	AN840-10R	.625	.500	AS5183R1008	AS5183R1008
AN840-10S		.625	.500	AS5183-1008	AS5183-1008
AN840-10T		.625	.500	None	
AN840-10W		.625	.500	AS5183W1008	
AN840-10WV		.625	.500		
AN840-11		.750	.375	AS5183B1206	
AN840-11D	AN840-11W	.750	.375	AS5183W1206	AS5183W1206
AN840-11DV	AN840-11WV	.750	.375	None	
None	AN840-11R	.750	.375	AS5183R1206	AS5183R1206
AN840-11S		.750	.375	AS5183-1206	AS5183-1206
AN840-11T		.750	.375	None	
AN840-11W		.750	.375	AS5183W1206	
AN840-11WV		.750	.375	None	

See note at end of table.

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TABLE III. Cross-reference data - Continued. 1/

Old AN PIN	Current AN PIN Inactive for new design	Tube Size	Pipe Size	Replacement AS PIN	Replacement AS PIN For new design
AN840-12		.750	.750	AS5183B1212	
AN840-12D	AN840-12W	.750	.750	AS5183W1212	AS5183W1212
AN840-12DV	AN840-12WV	.750	.750	None	
None	AN840-12R	.750	.750	AS5183R1212	AS5183R1212
AN840-12S		.750	.750	AS5183F-1212	AS5183-1212
AN840-12T		.750	.750	None	
AN840-12W		.750	.750	AS5183W1212	
AN840-12WV		.750	.750	None	
AN840-13		1.000	.375	AS5183B1606	
AN840-13D	AN840-13W	1.000	.375	AS5183W1606	AS5183W1606
AN840-13DV	AN840-13WV	1.000	.375	None	
None	AN840-13R	1.000	.375	AS5183R1606	AS5183R1606
AN840-13S		1.000	.375	AS5183-1606	AS5183-1606
AN840-13T		1.000	.375	None	
AN840-13W		1.000	.375	AS5183W1606	
AN840-13WV		1.000	.375	None	
AN840-16		1.000	.750	AS5183B1612	
AN840-16D	AN840-16W	1.000	.750	AS5183W1612	AS5183W1612
AN840-16DV	AN840-16WV				
None	AN840-16R	1.000	.750	AS5183R1612	AS5183R1612
AN840-16S		1.000	.750	AS5183F-1612	AS5183-1612
AN840-16T		1.000	.750	None	
AN840-16W		1.000	.750	AS5183W1612	
AN840-16WV		1.000	.750	None	
AN840-17		1.000	1.000	AS5183B1616	
AN840-17D	AN840-17W	1.000	1.000	AS5183W1616	AS5183W1616
AN840-17DV	AN840-17WV	1.000	1.000	None	
None	AN840-17R	1.000	1.000	AS5183R1616	AS5183R1616
AN840-17S		1.000	1.000	AS5183-1616	AS5183-1616
AN840-17T		1.000	1.000	None	
AN840-17W		1.000	1.000	AS5183W1616	
AN840-17WV		1.000	1.000	None	
AN840-20		1.250	1.250	AS5183B2020	
AN840-20D	AN840-20W	1.250	1.250	AS5183W2020	AS5183W2020
AN840-20DV	AN840-20WV	1.250	1.250	None	
None	AN840-20R	1.250	1.250	AS5183R2020	AS5183R2020
AN840-20S		1.250	1.250	AS5183-2020	AS5183-2020
AN840-20T		1.250	1.250	None	
AN840-20W		1.250	1.250	AS5183W2020	
AN840-20WV		1.250	1.250	None	
AN840-21		1.250	1.000	AS5183B2016	
AN840-21D	AN840-21W	1.250	1.000	AS5183W2016	AS5183W2016
AN840-21DV	AN840-21WV	1.250	1.000	None	
None	AN840-21R	1.250	1.000	AS5183R2016	AS5183R2016
AN840-21S		1.250	1.000	AS5183-2016	AS5183-2016
AN840-21T		1.250	1.000	None	
AN840-21W		1.250	1.000	AS5183W2016	
AN840-21WV		1.250	1.000	None	

See note at end of table.

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TABLE III. Cross-reference data - Continued. 1/

Old AN PIN	Current AN PIN Inactive for new design	Tube Size	Pipe Size	Replacement AS PIN	Replacement AS PIN For new design
AN840-24		1.500	1.500	AS5183B2424	
AN840-24D	AN840-24W	1.500	1.500	AS5183W2424	AS5183W2424
AN840-24DV	AN840-24WV	1.500	1.500	None	
None	AN840-24R	1.500	1.500	AS5183R2424	AS5183R2424
AN840-24S		1.500	1.500	AS5183-2424	AS5183-2424
AN840-24T		1.500	1.500	None	12581258
AN840-24W		1.500	1.500	AS5183W2424	
AN840-24WV		1.500	1.500	None	
AN840-25		1.500	1.250	AS5183B2420	
AN840-25D	AN840-25W	1.500	1.250	AS5183W2420	AS5183W2420
AN840-25DV	AN840-25WV	1.500	1.250	None	
None	AN840-25R	1.500	1.250	AS5183R2420	AS5183R2420
AN840-25S		1.500	1.250	AS5183-2420	AS5183-2420
AN840-25T		1.500	1.250	None	
AN840-25W		1.500	1.250	AS5183W2420	
AN840-25WV		1.500	1.250	None	

1/ For new design use material designator R and W.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to SAE-AS4843/2, this document references the following:

SAE AMS-STD-595	SAE-AMS-QQ-P-416	SAE-AMS4614
MIL-PRF-8625	SAE-AMS-QQ-S-763	SAE-AMS5639
MIL-DTL-83488	SAE-AMS-S-6758	SAE-AMS5645
MS21344	SAE-AMS-6370	SAE-AMS5648
ASME B46.1	SAE-AMS4124	SAE-AMS6382
ASTM B124/B124M	SAE-AMS4133	SAE-ARP1590
ASTM B138/B138M	SAE-AMS4141	SAE-AS611
SAE-AMS-QQ-A-225/6	SAE-AMS2472	SAE-AS5132
SAE-AMS-QQ-A-225/9	SAE-AMS2700	SAE-AS5183
SAE-AMS-QQ-A-367	SAE-AMS4339	SAE-AS5201

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force – 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2025-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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Check the source to verify that this is the current version before use.